PHÁT TRIỂN ỨNG DỤNG DI ĐỘNG (Tài liệu hướng dẫn tuần 10)

Google play service

Add permission (AndroidManifest.xml)

```
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
<uses-permission android:name ="android.permission.ACCESS_COARSE_LOCATION"/>
```

Add dependencies (build.gradle)

```
implementation("com.google.android.gms:play-services-maps:18.1.0")
implementation("com.google.android.gms:play-services-location:21.3.0")
```

Define API Key (inside “application” tag in AndroidManifest.xml)

```
<meta-data android:name="com.google.android.geo.API_KEY"

           android:value="YOUR API KEY"/>
```

Call API

XML Layout

```
<fragment
    android:id="@+id/map"
    android:name="com.google.android.gms.maps.SupportMapFragment"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity" />
```

FusedLocationProviderClient

```
private GoogleMap mMap;
private FusedLocationProviderClient fusedLocationClient;


onCreate:

SupportMapFragment mapFragment = (SupportMapFragment)
          getSupportFragmentManager().findFragmentById(R.id.map);

mapFragment.getMapAsync(this);

fusedLocationClient =
                LocationServices.getFusedLocationProviderClient(this);
```

Runtime request permission

```
@Override
public void onMapReady(@NonNull GoogleMap googleMap) {
    mMap = googleMap;

    // Check and request location permissions


    // Call request location updates
}
```

Request location update(latitude, longitude, ...)

```
LocationRequest.Builder builder = new LocationRequest.Builder(5000);
builder.setPriority(Priority.PRIORITY_HIGH_ACCURACY);
LocationRequest locationRequest = builder.build();

fusedLocationClient.requestLocationUpdates(locationRequest, new LocationCallback() {
    @Override
    public void onLocationResult(@NonNull LocationResult locationResult) {
        ...
    }
}, Looper.getMainLooper());
```

From location to address

```
Location location = locationResult.getLastLocation();
if (location != null) {
    LatLng currentLocation = new LatLng(location.getLatitude(),
                                        location.getLongitude());
    Log.d("MapDebug", "Latitude: " + location.getLatitude()
                                   + ", Longitude: "
                                   + location.getLongitude());
    mMap.clear();
    mMap.addMarker(new MarkerOptions()
                           .position(currentLocation)
                           .title(getAddress(location)));
    mMap.moveCamera(
          CameraUpdateFactory.newLatLngZoom(currentLocation, 20));
}
```

From location to address

```
private String getAddress(Location location) {
    Geocoder geocoder = new Geocoder(this, Locale.getDefault());
    try {
        List<Address> addresses =
                geocoder.getFromLocation(location.getLatitude(),
                                         location.getLongitude(), 1);
        if (addresses != null && !addresses.isEmpty()) {
            return addresses.get(0).getAddressLine(0);
        }
    } catch (IOException e) {
        e.printStackTrace();
    }
    return "Unknown Location";
}
```

Geocoding address into coordinates

```
List<Address> addresses = geocoder.getFromLocationName(
        "19 Nguyen Huu Tho, Tan Phong, Quan 7, Ho Chi Minh, Viet Nam",
        1);

Address firstAddress = addresses.get(0);
double latitude = firstAddress.getLatitude();
```

```
double longitude = firstAddress.getLongitude();
Log.d("Geocoding", latitude + " - " + longitude);
```